

SOA and RPC

Service-Oriented Architecture and RPC

Learning Objectives

Be able to..

- describe Service-Oriented Architecture (SOA) and microservices and how they relate to client-server systems.
- define Remote Procedure Call (RPC), what stubs/skeletons do, and the problems that they solve.
- reason about RPC semantics and understand how to implement them.

Service-Oriented Architecture & Microservices

- At internet-scale, large scale systems require more than just one software system!
 - Scalability
 - Fault tolerance
 - Development organization
- Service Oriented Architecture
 - split an application into services that each expose a network API.
 - Each service scaled, has fault managed, and is developed independently.
 - Today's systems often use "microservices": super small services!
- Each service is both a client and a server
 - Client because they invoke other services
 - Server because they respond to other services
- Huge ecosystem of new challenges/solutions in this space (load balancers, API gateways, service discovery, versioning issues, consistency, partial failures, etc.)

Remote Procedure Class (RPC)